System Administration and Maintenance

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2. Communication Concepts

- Internet Protocol Suite
- TCP/IP
- DNS
- FTP
- SMTP

Learning Outcomes

By the end of this lecture, students should be able to Describe fundamental concepts on computer networks :

- Explain the Internet Protocol Suite.
- Describe the TCP/IP model and its layers.
- Differentiate TCP and IP.
- Explain how DNS translates names to IP addresses.
- Describe FTP file transfer operations.
- Explain how SMTP delivers email messages.
- Identify real-world uses of these protocols.

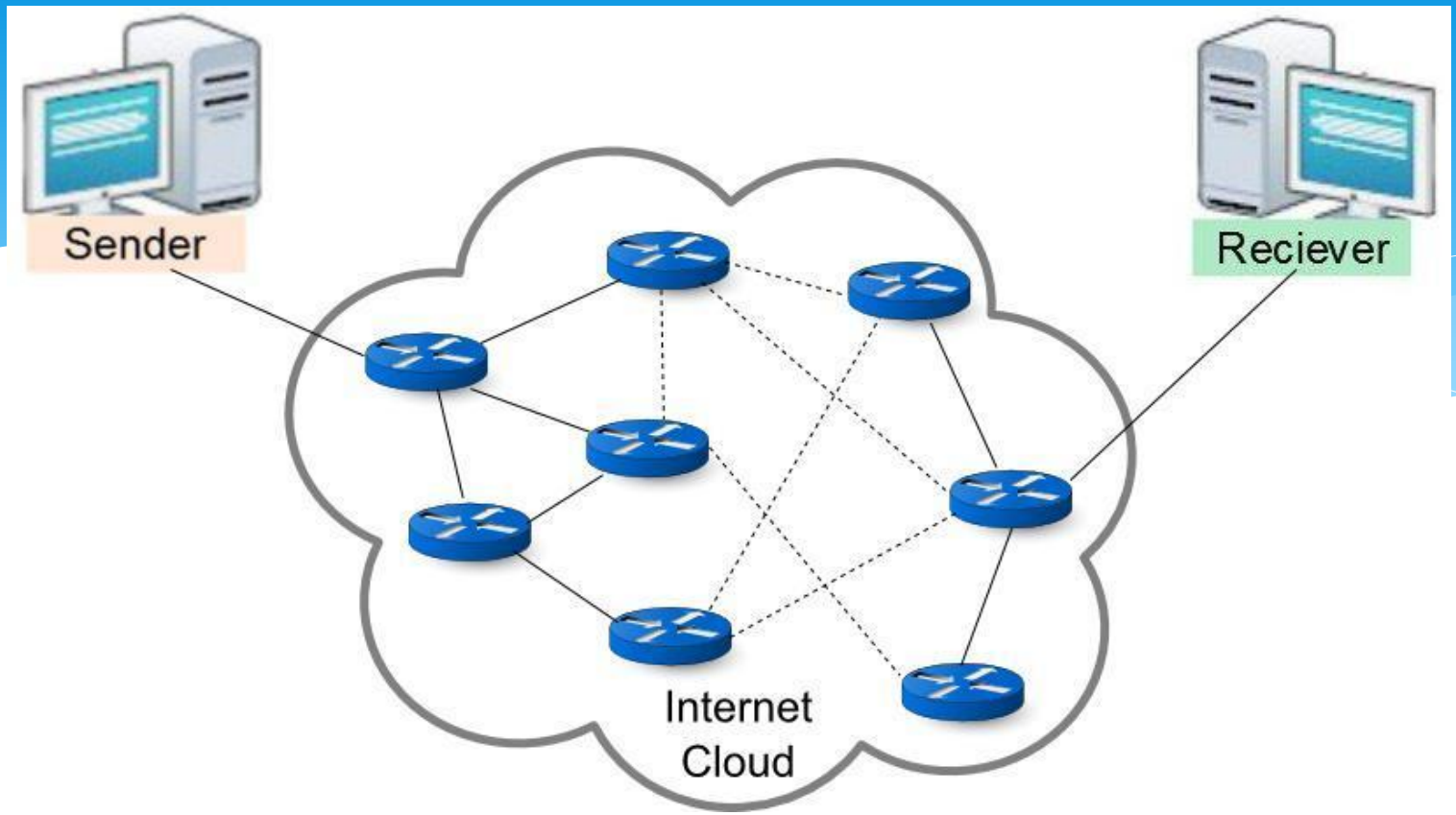
What Is Network Communication

* Computer communication is the exchange of data between computers through communication media.

* **Components**
  - Sender / Source
  - Receiver / Destination
  - Transmission Medium
  - Protocol
  - Message

* **Example**

  Laptop → Router → Internet → Web Server



- Communication = Exchange of data between devices.
- Devices communicate using protocols.
- A protocol defines rules, format, timing, and error handling.
- The Internet works because millions of devices follow the same protocol suite.

What is a Protocol

A protocol is a set of rules that governs communication between devices.

* Protocols define
  - Data format
  - Transmission speed
  - Error detection
  - Synchronization
  - Security

* Eg:
  - TCP
  - IP
  - HTTP
  - HTTPS
  - FTP
  - SMTP

Internet Protocol Suite

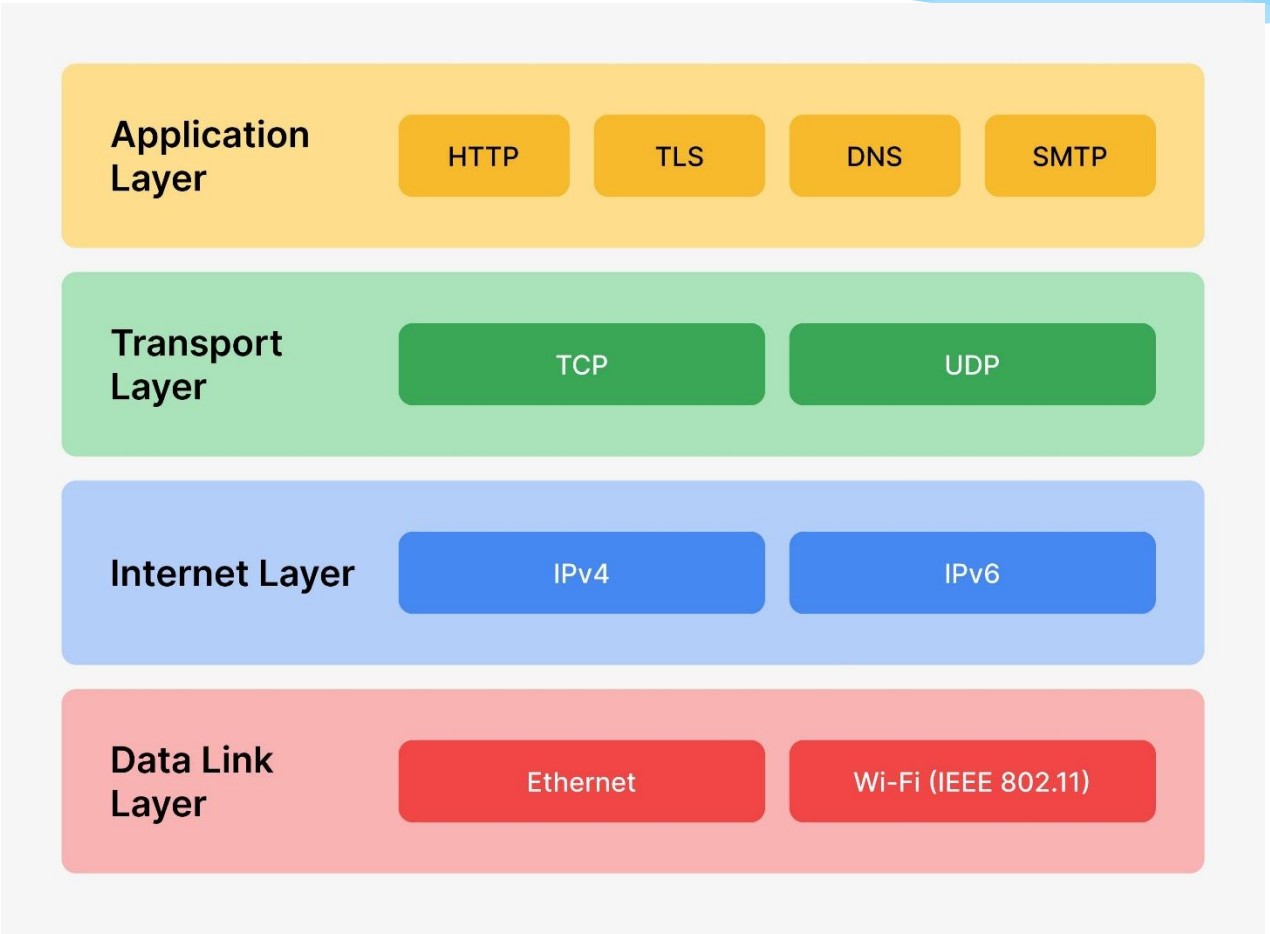
The Internet Protocol Suite is the collection of communication protocols used to connect devices over the Internet.

Also known as **TCP/IP Protocol Suite**

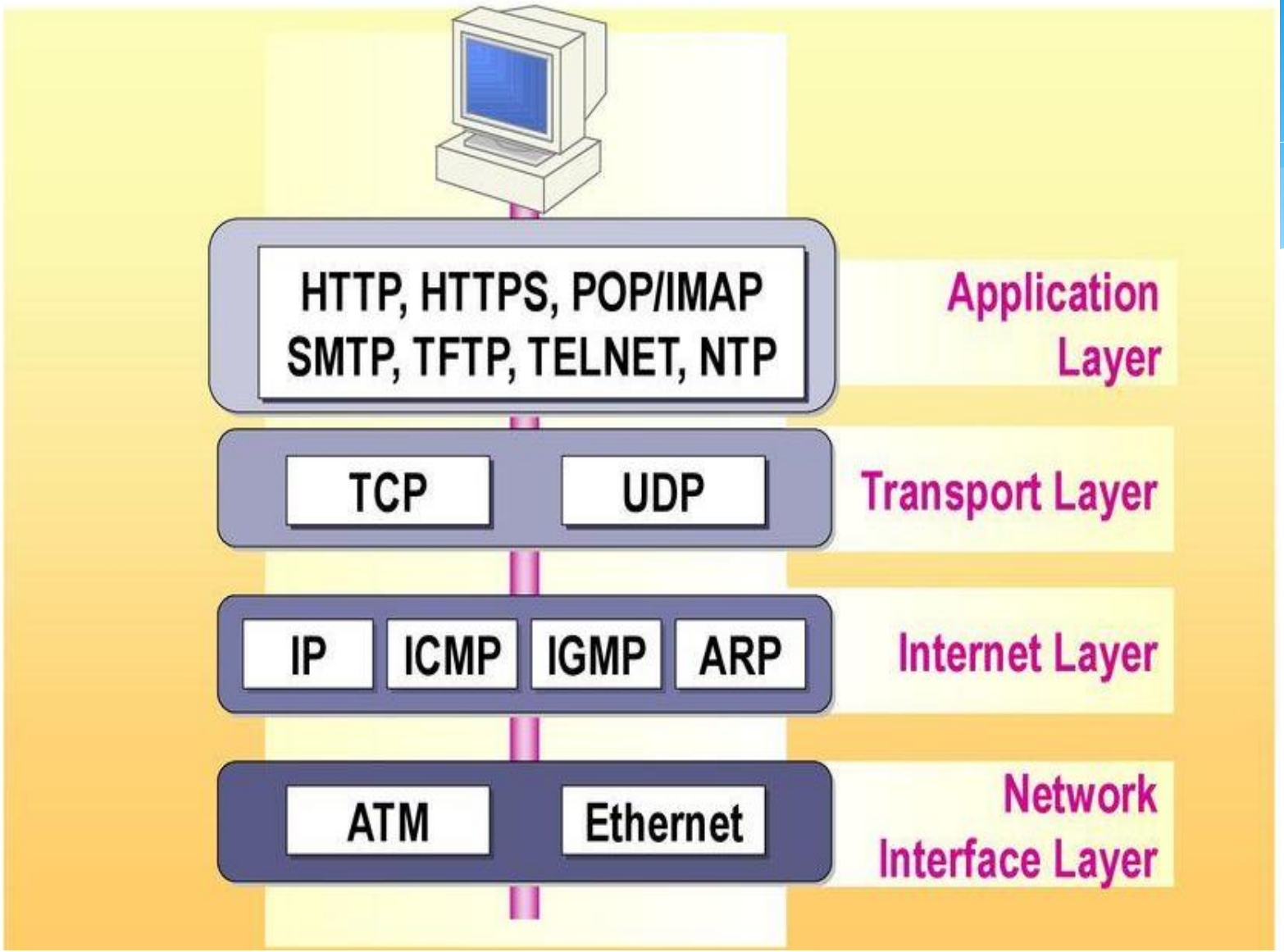
* It allows
  - Data communication
  - Internet browsing
  - Email
  - File sharing
  - Video conferencing

Internet Protocol Suite

A collection of communication protocols used on the Internet.

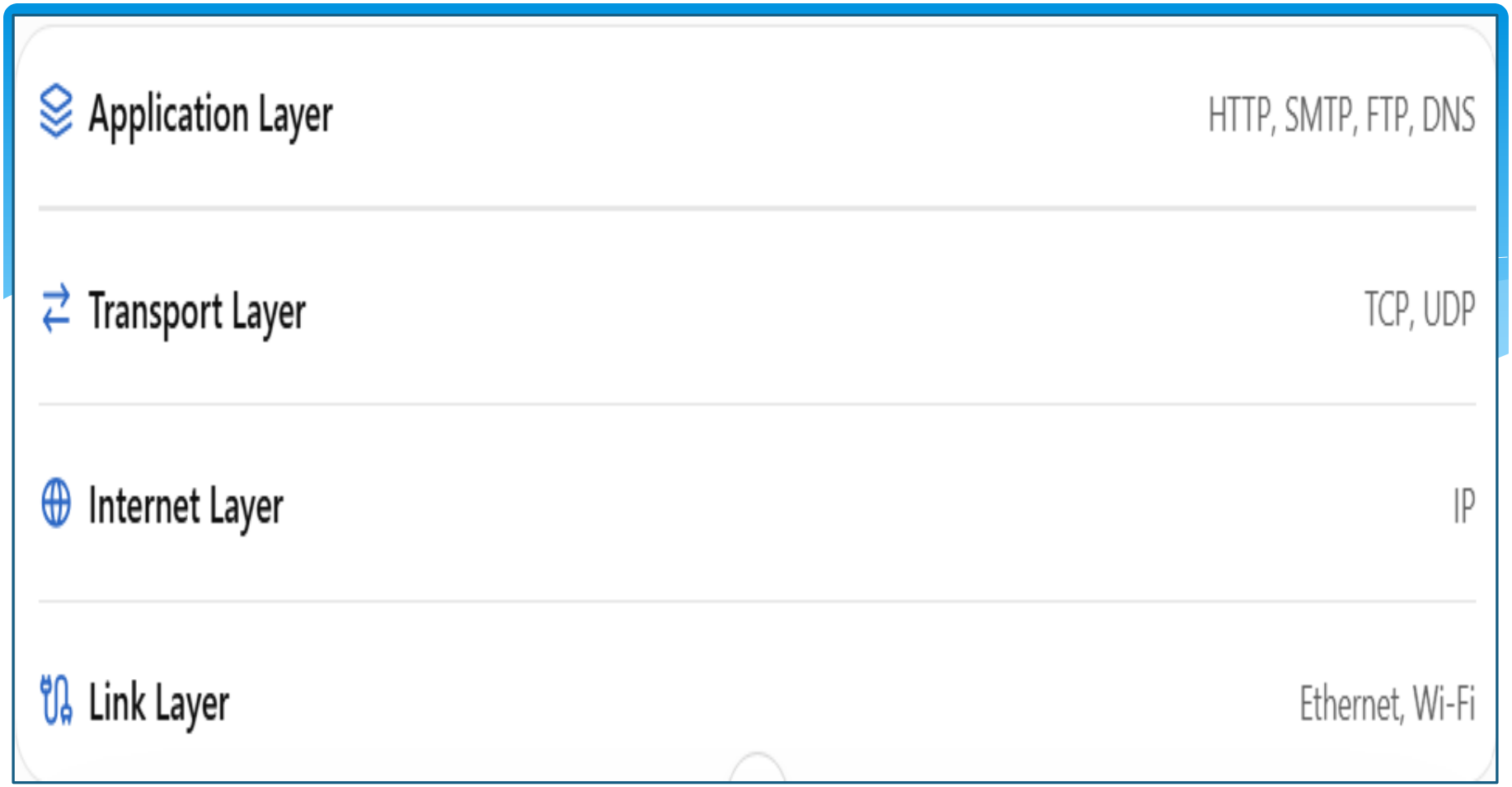


THE TCP/IP PROTOCOL SUITE



TCP/IP Architecture

Layer	Function	Example Protocols
Application	User Services	HTTP, FTP, SMTP, DNS
Transport	End-to-End Communication	TCP, UDP
Internet	Addressing & Routing	IP
Link	Physical Transmission	Ethernet, Wi-Fi



Each layer performs a specific function and passes data to the next layer.

Transmission Control Protocol (TCP)

* TCP provides
  - Reliable communication
  - Error checking
  - Flow control
  - Data sequencing

* Characteristics
  - Connection-oriented
  - Reliable
  - Ordered delivery

* Applications
  - Email
  - Web browsing
  - Online banking

Internet Protocol (IP)

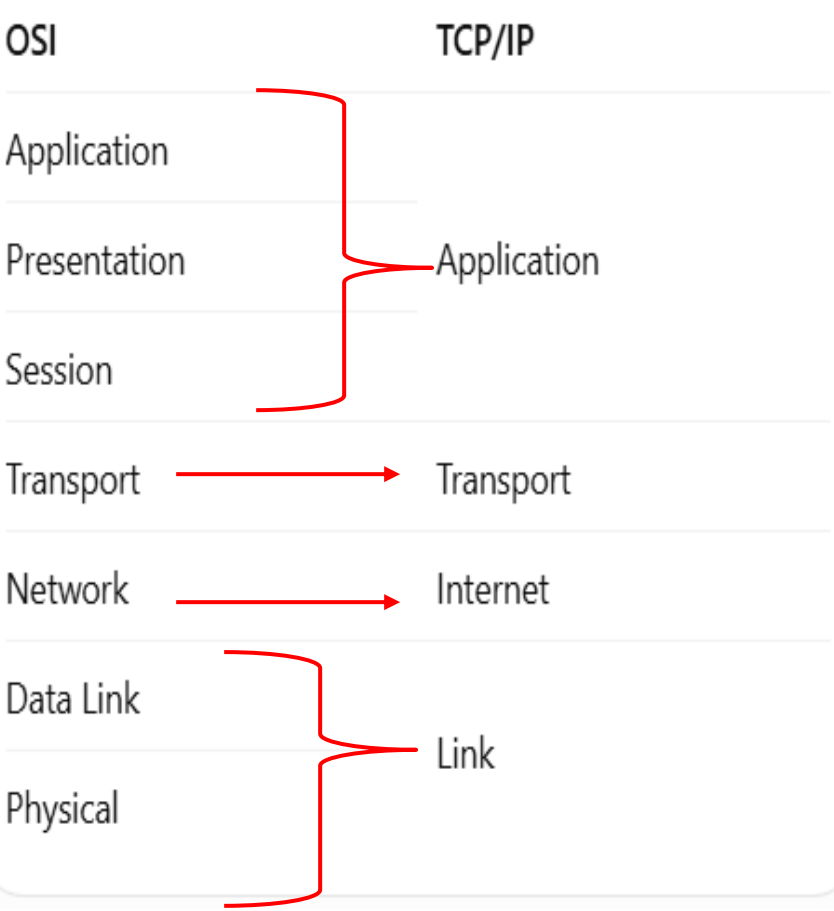
* IP is responsible for
  - Logical addressing
  - Packet routing
  - Delivering packets
* IPv4 Example
  - 192.168.1.100
* IPv6 Example
  - 2001:db8::1
* Functions
  - Address assignment
  - Routing
  - Fragmentation

TCP/IP Model

TCP/IP Layer

Application	User services
Transport	End-to-end delivery
Internet	Routing
Link	Physical transmission

OSI Mapping



TCP vs IP

IP = postal system (finding the address)

TCP = registered mail service (ensures delivery)

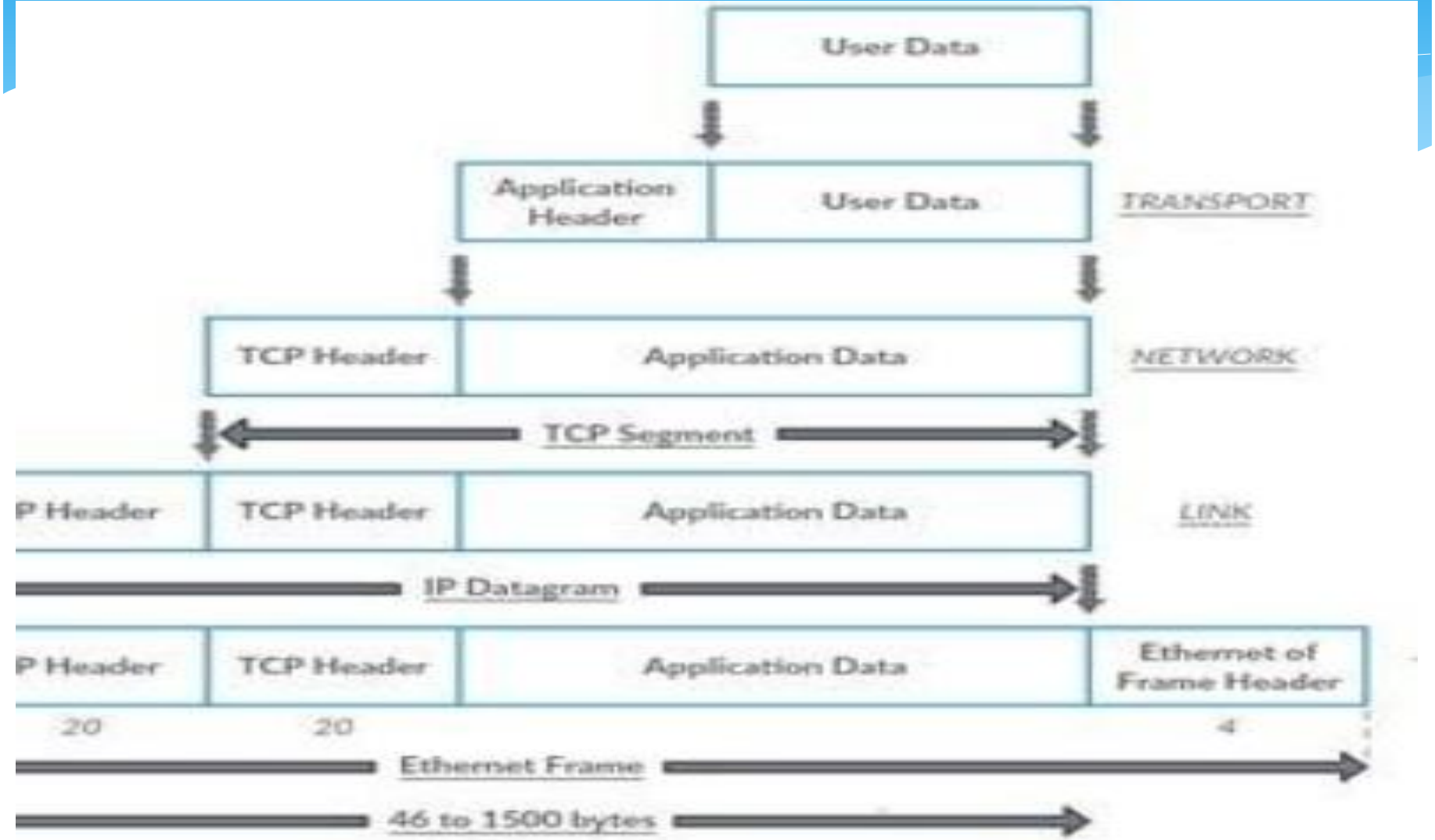
Feature	TCP	IP
Layer	Transport	Internet
Purpose	Reliable delivery	Addressing & routing
Connection	Connection-oriented	Connectionless
Error checking	Yes	Limited
Example	Web page download	Routing packets across Internet

TCP vs IP

TCP	IP
Reliable	Best effort
Connection-oriented	Connectionless
Error checking	Routing
Data sequencing	Packet delivery

Together TCP + IP = Reliable Internet Communication

How TCP/IP Works



* Encapsulation process:
  - Application creates data.
  - TCP adds a TCP header.
  - IP adds an IP header.
  - Link layer adds frame information.
  - Bits travel across the network.

* At the destination: the process is reversed (decapsulation).

Domain Name System (DNS)

* DNS translates **Domain Names** into **IP Addresses**

* Example :

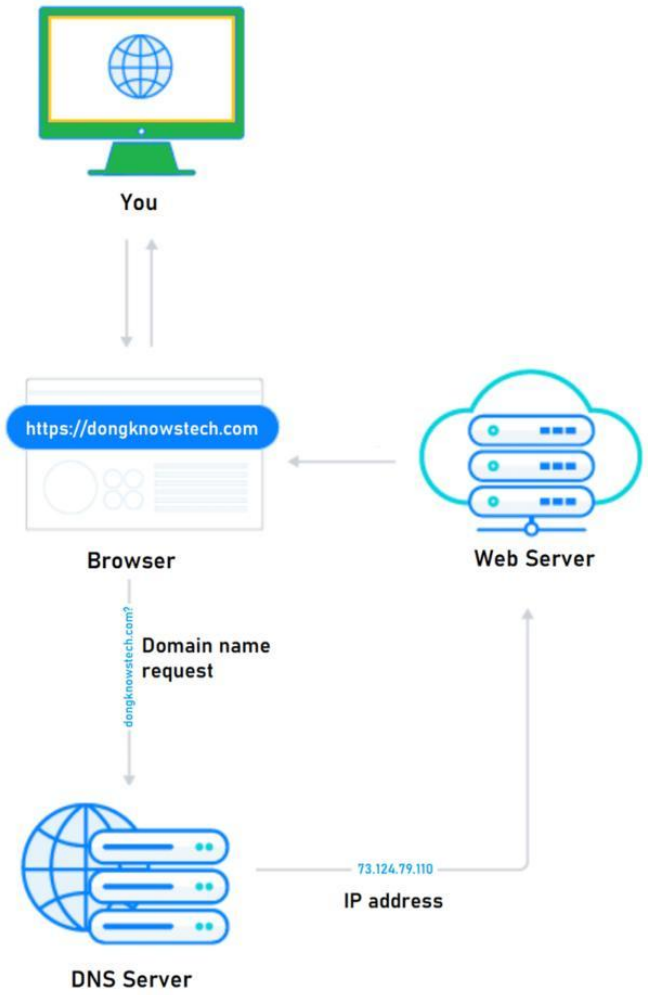
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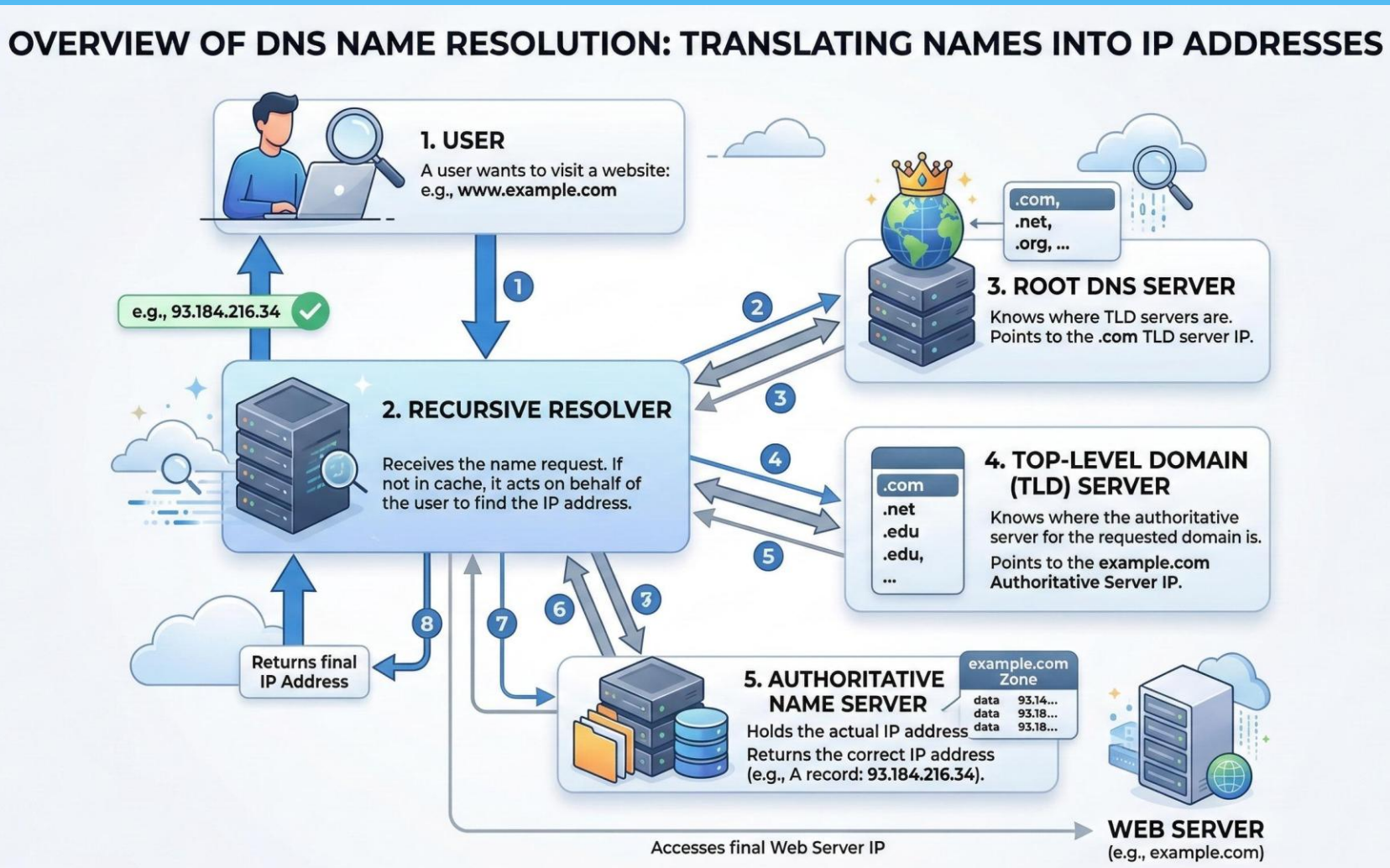
* Without DNS users would need to remember IP addresses.

DNS (Domain Name System)



Purpose: Converts domain names into IP addresses.

DNS (Domain Name System)



DNS Resolution Process

User enters website name

↓

Browser queries DNS server

↓

DNS finds IP address

↓

Browser contacts web server

↓

Website loads

DNS Lookup Steps

1. User enters www.example.com.
2. Browser asks the DNS resolver.
3. Resolver contacts DNS servers.
4. IP address is returned.
5. Browser connects to the web server.

* Without DNS, users would need to remember numerical IP addresses.

File Transfer Protocol (FTP)

* FTP is used to Transfer files between computers over a network.

* Common Uses
  - Website uploading
  - Downloading files
  - Data backup
  - File sharing

* Default Port : **21**

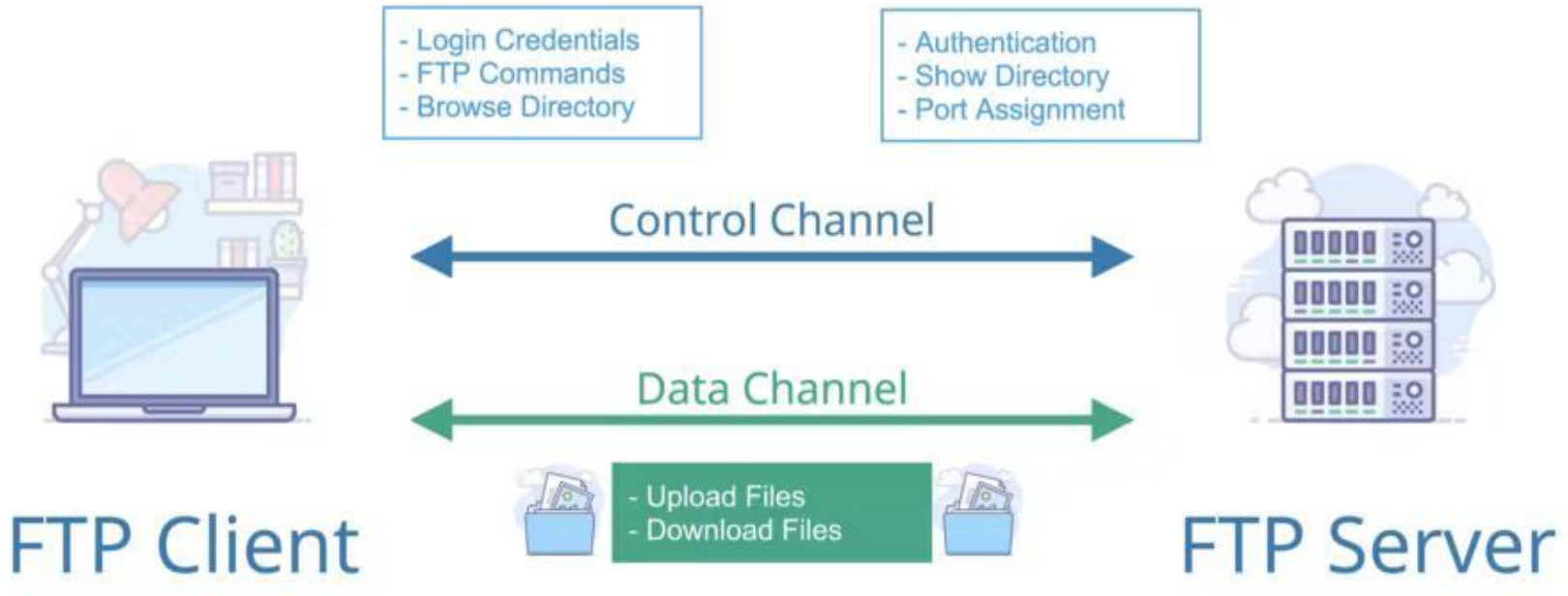
FTP Operation

- FTP Process:

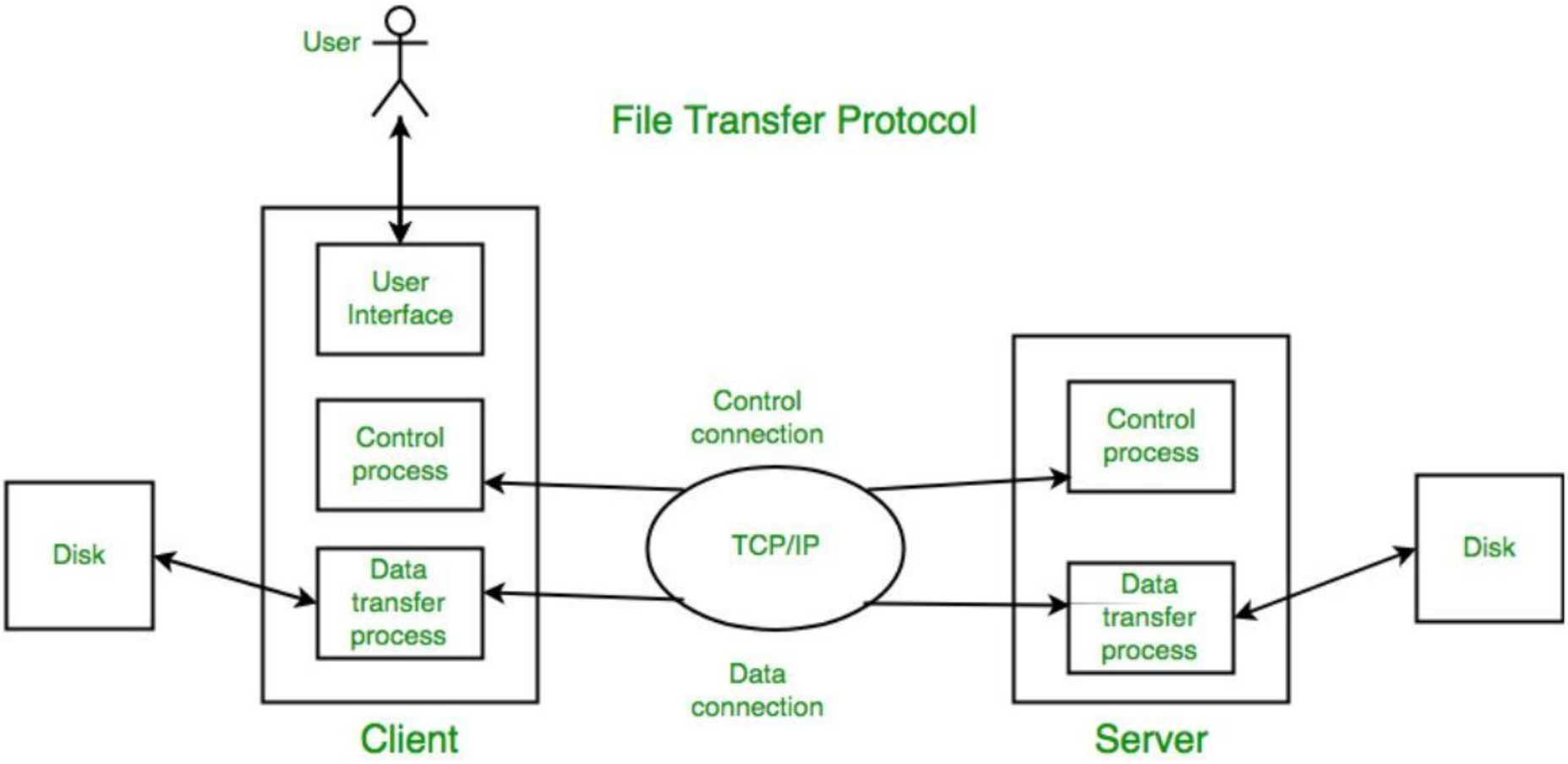
Client
↓
Connects to FTP Server
↓
Login
↓
Upload/Download Files
↓
Disconnect

- Popular FTP Software
  - FileZilla
  - WinSCP
  - Cyberduck

FTP (File Transfer Protocol)



Purpose: Transfer files between a client and a server.


User
File Transfer Protocol
User Interface
Control process
Data transfer process
Disk
Client
Control connection
TCP/IP
Data connection
Control process
Data transfer process
Disk
Server

Common Operations

- Upload files
- Download files
- Rename files
- Delete files
- Create folders

Security Note

Traditional FTP sends **usernames and passwords in plain text.**

Safer alternatives:

- **SFTP** (SSH File Transfer Protocol)
- **FTPS** (FTP over TLS/SSL)

Simple Mail Transfer Protocol (SMTP)

* SMTP is the standard protocol used to send emails.
* Functions
  - Send outgoing emails
  - Transfer emails between mail servers

* Default Ports
  - 25
  - 465
  - 587

Email Communication Process

Sender

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SMTP Server

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Internet

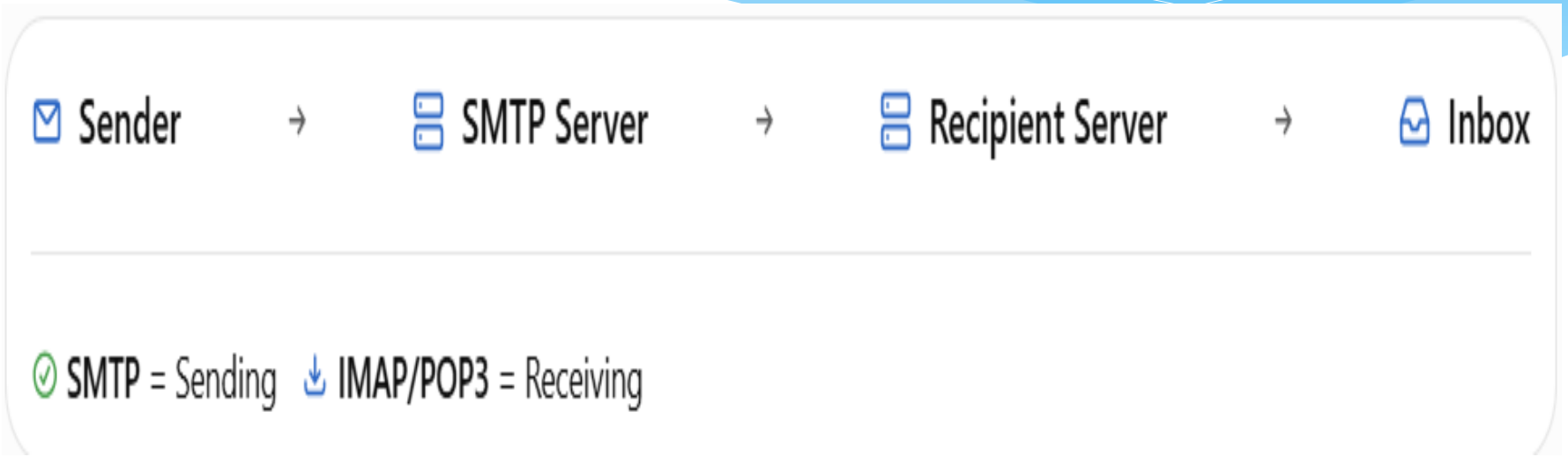
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Recipient Mail Server

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Receiver

- Receiving protocols
  - POP3
  - IMAP

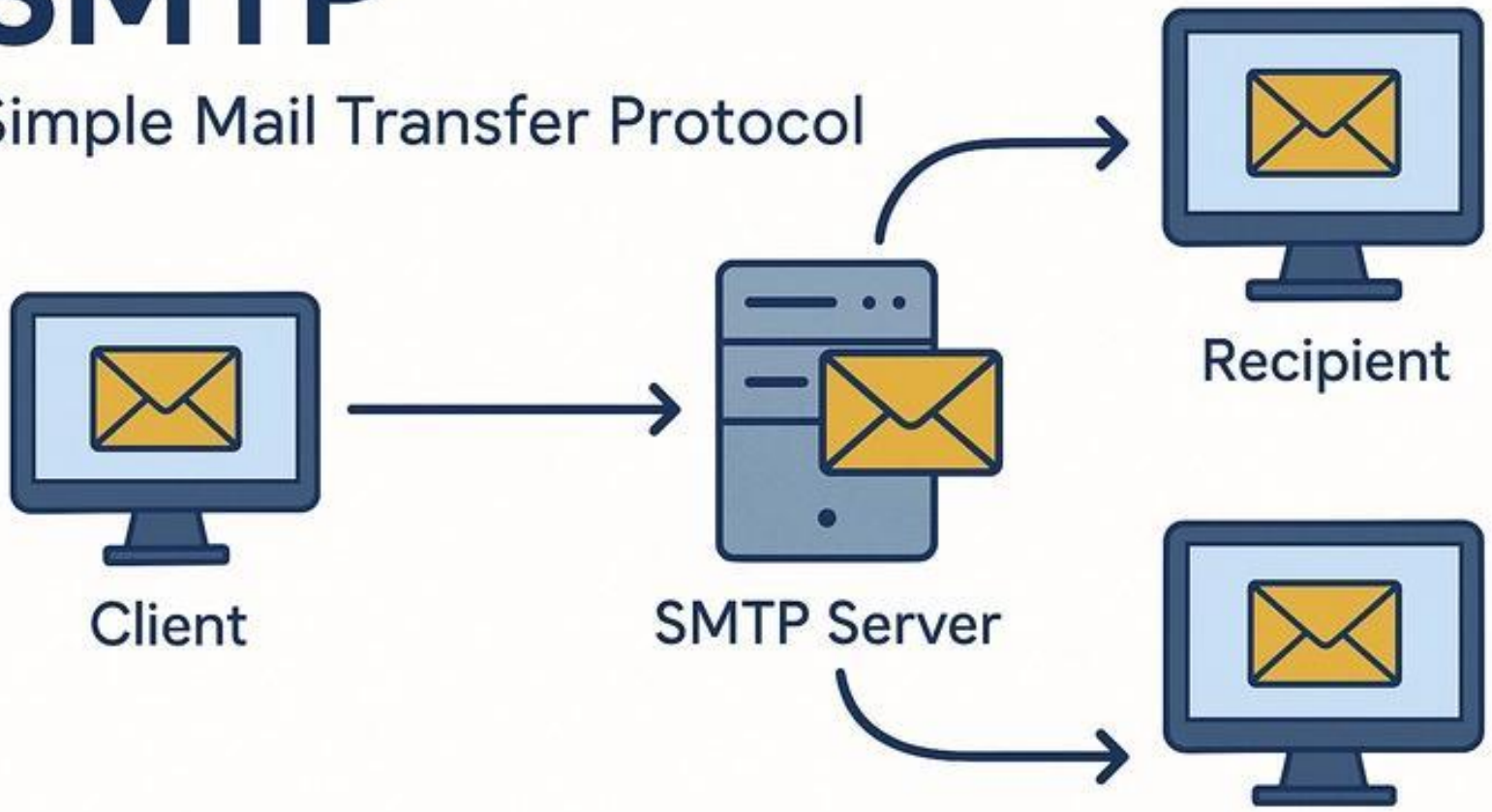


* Eg:

  When you send a Gmail message, SMTP transfers it to the recipient's mail server.

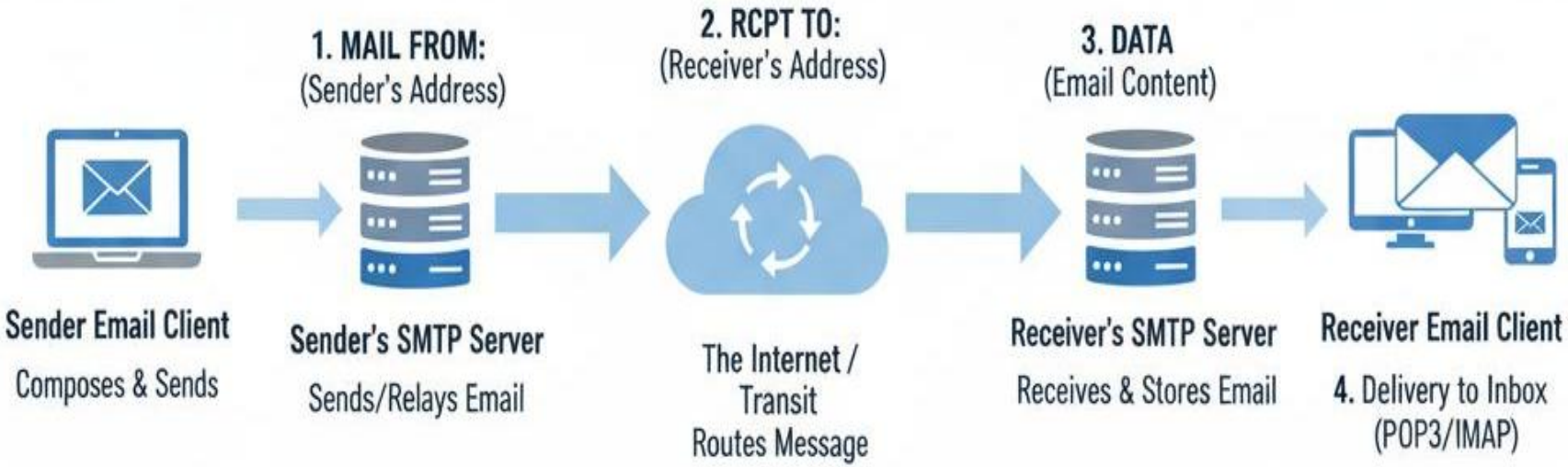
SMTP

Simple Mail Transfer Protocol



Purpose: Send email messages across networks.

SMTP EMAIL FLOW

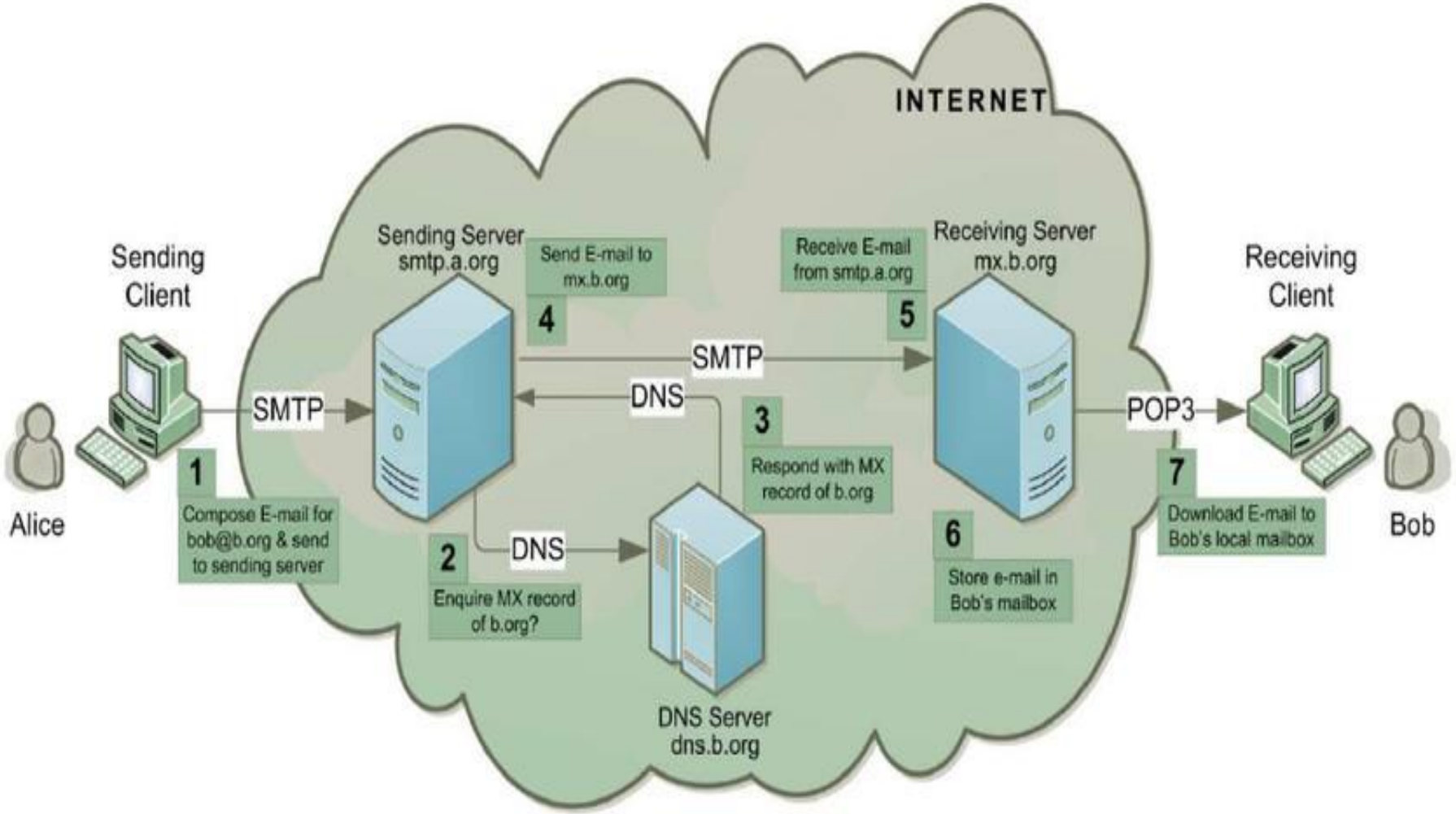


Comparison of Protocols

Protocol	Purpose	Default Port
TCP	Reliable Transmission	-
IP	Addressing	-
DNS	Name Resolution	53
FTP	File Transfer	21
SMTP	Email Sending	25, 465, 587

Comparison of Protocols

Protocol	Purpose	Layer
IP	Addressing & routing	Internet
TCP	Reliable delivery	Transport
DNS	Name resolution	Application
FTP	File transfer	Application
SMTP	Email sending	Application


INTERNET
Sending Client
Sending Server
smtp.a.org
Send E-mail to
mx.b.org
4
Receive E-mail
from smtp.a.org
5
Receiving Server
mx.b.org
Receiving Client
SMTP
SMTP
DNS
3
Respond with MX
record of b.org
POP3
Alice
1
Compose E-mail for
bob@b.org & send
to sending server
2
DNS
Enquire MX record
of b.org?
6
Store e-mail in
Bob's mailbox
7
Download E-mail to
Bob's local mailbox
Bob
DNS Server
dns.b.org

Real-Life Applications

* **TCP/IP -** Internet browsing
* **DNS -** Finding websites
* **FTP -** Website management
* **SMTP -** Email communication
* **Used By**
  - Google
  - Microsoft
  - Universities
  - Banks
  - Government organizations

Real-World Example

What happens when a student opens www.kdu.ac.lk?

1. DNS — Finds IP address

Browser asks DNS: "What is the IP for **www.kdu.ac.lk**?"

2. TCP — Creates connection

Browser establishes a reliable connection with the server.

3. IP — Routes packets

Packets travel through routers across the Internet.

4. Application — Transfers webpage

HTTP transfers the webpage back to the browser.

Result: The website appears on the student's screen.

Advantages Internet Protocol Suite

- Standardized
- Reliable
- Scalable
- Interoperable
- Supports millions of devices
- Global communication

Quick Quiz

- Which protocol translates google.com into an IP address?
- Which protocol ensures reliable delivery?
- Which protocol sends email?
- Which protocol transfers files?
- Which layer performs routing?

THANK YOU